

Neural Particle Method: Experimental Record

Synthetic-data setup, machinery, and the benchmark suite

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Abstract

This note records, in one place, the synthetic markets and dynamics used throughout the project, the simulation, estimation and scoring machinery, and the specification of the clean benchmark suite agreed on 8 September 2026 (Section 4), run on 9 September 2026; its tables are generated from the MLflow store. The suite follows the discovery, in the diagnostics of 8 September, that all one-pass schemes were fitting the conditional expectation of the raw Euler variance rather than of its positive part, which the dynamics actually use (Section 3.4); the diagnostic runs themselves live in the MLflow experiments `mechanism`, `bump_floor` and `bump_floor_nn`.

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1 Conventions

Explicit denotes the one-pass, per-slice calibration with a per-slice neural network. Nadaraya–Watson (NW) is the classical particle method: the same one-pass structure with kernel regression. Other per-slice estimators are named by their estimator (RKHS, spline, ridge). *Implicit* denotes the damped outer iteration with a single network in (t, x) . All errors are implied-volatility root-mean-square errors in volatility basis points against the target surface, unless stated otherwise. “Pooled” means over all quoted strikes and maturities together; “wings” means the quoted strikes with $|k| > 0.25$; per-maturity columns restrict to one maturity.

2 Synthetic data

2.1 Target surfaces: SSVI power law

Each scenario’s market is an SSVI implied-variance surface in the power-law parameterisation. With log-moneyness $k = \log(K/S_0)$, $S_0 = 1$, and parameters $(\sigma_0, \eta, \gamma, \rho_s)$,

$$\theta_T = \sigma_0^2 T, \quad \varphi(\theta_T) = \eta \theta_T^{-\gamma}, \quad w(k, T) = \frac{1}{2} \theta_T \left(1 + \rho_s \varphi k + \sqrt{(\varphi k + \rho_s)^2 + 1 - \rho_s^2} \right), \quad (1)$$

where w is total implied variance and the implied volatility is $\sqrt{w(k, T)/T}$. Parameters are drawn independently per scenario from

$$\sigma_0 \sim U(0.12, 0.35), \quad \eta \sim U(0.5, 2.0), \quad \gamma \sim U(0.3, 0.5), \quad \rho_s \sim U(-0.85, -0.10),$$

resampled until the sufficient no-arbitrage conditions $0 < \gamma \leq \frac{1}{2}$, $|\rho_s| < 1$, $\eta(1 + |\rho_s|) \leq 2$ hold. Scenario s_i uses the seed $1000 + i$ (`bench/scenarios.py`, `make_registry`).

The Dupire local volatility is analytic from Equation (1): with w_k, w_{kk}, w_T the derivatives of w ,

$$\sigma_{\text{Dup}}^2(T, k) = \frac{w_T}{1 - \frac{k}{w} w_k + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) w_k^2 + \frac{1}{2} w_{kk}}, \quad (2)$$

with the denominator floored at 0.05, the local variance clipped to $[10^{-8}, 9]$, and the time argument clipped to $[t_{\min}, T]$ with $t_{\min} = 0.004$ (`market/local_vol.py`). These clipping constants are part of the calibrated behaviour and are never changed.

2.2 Variance dynamics

The calibrated model is Heston-type LSV in log-spot $X = \log S$,

$$dX_t = -\frac{1}{2} L^2(t, X_t) V_t dt + L(t, X_t) \sqrt{V_t} dW_t, \quad dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dB_t, \quad d\langle W, B \rangle_t = \rho dt. \quad (3)$$

For scenario s_i the dynamics parameters are drawn from the same seed as the surface: $\kappa \in \{1, 2, 3\}$, $\xi \in \{0.3, 0.6, 1.0\}$, $\rho \in \{-0.7, -0.3\}$, each uniform over its set, and $\theta = v_0 = \sigma_0^2$ so that the variance level matches the surface’s short-end level. The Feller ratio $2\kappa\theta/\xi^2$ therefore ranges from 0.05 to 4.3 across the grid. Table 1 lists all twenty scenarios. Horizon $T = 2$ years; spot $S_0 = 1$; no rates or dividends.

Two further families exist in the registry and were not used today: six variants of s01 with (ξ, ρ) swept over the grid (`fig3_registry`), and three Heston-market scenarios whose target surface is generated by a Heston model with parameters different from the calibrated dynamics (`li_simple`, `li_complex`, `bayer`; horizon 1 year; Dupire surface from prices on a 50×121 grid).

2.3 Quotes, bump, and scoring

Quoted strikes are 13 log-strikes equally spaced in log between 0.6 and 1.6; maturities are 0.25, 0.5, 1 and 2 years. The wings are the strikes with $|k| > 0.25$ (six of thirteen). The warm-start

sid	SSVI surface				dynamics			Feller
	σ_0	η	γ	ρ_s	κ	ξ	ρ	
s01	0.261	0.52	0.34	-0.21	1	0.3	-0.7	1.51
s02	0.208	1.04	0.45	-0.56	3	0.6	-0.7	0.72
s03	0.163	0.89	0.40	-0.35	3	1.0	-0.7	0.16
s04	0.120	0.70	0.33	-0.56	2	0.6	-0.3	0.16
s05	0.285	1.01	0.49	-0.72	2	1.0	-0.3	0.32
s06	0.304	1.00	0.38	-0.58	3	1.0	-0.3	0.55
s07	0.202	0.56	0.46	-0.49	3	1.0	-0.3	0.25
s08	0.158	1.72	0.34	-0.11	1	1.0	-0.7	0.05
s09	0.133	0.99	0.31	-0.64	3	1.0	-0.3	0.11
s10	0.142	1.30	0.48	-0.11	3	1.0	-0.7	0.12
s11	0.144	1.10	0.38	-0.11	2	1.0	-0.7	0.08
s12	0.191	1.01	0.33	-0.22	3	1.0	-0.7	0.22
s13	0.303	0.62	0.43	-0.51	2	1.0	-0.3	0.37
s14	0.220	1.38	0.48	-0.11	1	0.6	-0.3	0.27
s15	0.127	0.72	0.37	-0.33	3	0.3	-0.3	1.08
s16	0.172	1.16	0.37	-0.42	1	0.6	-0.7	0.17
s17	0.136	1.13	0.42	-0.50	1	0.3	-0.3	0.41
s18	0.257	1.18	0.47	-0.24	3	0.6	-0.3	1.10
s19	0.170	1.09	0.39	-0.23	2	0.3	-0.7	1.28
s20	0.253	0.93	0.32	-0.32	3	0.3	-0.7	4.27

Table 1: The twenty synthetic scenarios: SSVI surface parameters, variance dynamics, and the Feller ratio $2\kappa\theta/\xi^2$.

experiments bump the surface to $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, a level rise of one volatility point with curvature and skew nudged inside the no-arbitrage region.

Scoring reprices vanilla calls under the calibrated leverage field on fresh paths: 500 000 particles, 200 Euler steps over $[0, 2]$, seed 10 000 plus a strategy-specific offset, call price as the sample mean of the payoff, Black–Scholes inversion at the grid-snapped maturity, and the exact SSVI implied volatility at the same snapped time as the target (`pricing/reprice.py`, `pricing/metrics.py`). The bump experiments use 300 000 repricing paths. Errors are reported as RMSE in volatility bp, pooled, wings-only, and per maturity.

3 Machinery

3.1 Simulation

Calibration and repricing share one stepper. With $\Delta = T/K$, $v^+ = \max(v, 0)$, $z_1 = \rho z_b + \sqrt{1 - \rho^2} z_p$ and independent standard normals z_b, z_p ,

$$x_{k+1} = x_k - \frac{1}{2}L_k(x_k)^2 v_k^+ \Delta + L_k(x_k) \sqrt{v_k^+} \sqrt{\Delta} z_1, \quad v_{k+1} = v_k + \kappa(\theta - v_k^+) \Delta + \xi \sqrt{v_k^+} \sqrt{\Delta} z_b. \quad (4)$$

This is full truncation: the raw v may be negative and is carried, but only v^+ enters any coefficient. The paper calibrated with $K = 50$ steps; the benchmark suite of Section 4 uses 200 for calibration and repricing alike. The leverage is stored per slice on a grid in log-spot and evaluated by linear interpolation, constant beyond the grid ends; the slice in force at time t is the last grid time $\leq t$. The explicit and NW passes use a per-slice grid at the 0.1% to 99.9% quantiles of the current cloud (101 points); the implicit loop uses a fixed grid of 81 points from log 0.4 to log 2.2. The leverage is capped at $L_{\max} = 4$ and the fitted conditional expectation is floored at 10^{-4} .

3.2 One-pass schemes and their estimators

At slice k , fit $f_k(x) \approx \mathbb{E}[V_{t_k} \mid X_{t_k} = x]$ on a subsample of 30 000 particles from the current cloud (the earlier default of the code; the benchmark suite of Section 4 fits on the full cloud), set $L_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\text{max}})$, and advance the cloud. Slice 0 uses $f_0 \equiv v_0$. Estimators:

- **NW**: Gaussian Nadaraya–Watson with Silverman bandwidth $h = 1.06 \hat{\sigma}_x n^{-1/5}$ on the subsample.
- **Explicit (per-slice network)**: a $1 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network with softplus output scaled by 0.04, input scaled by 0.3, trained by Adam at learning rate 10^{-2} on the unweighted squared loss; 400 steps on the first fitted slice and 120 steps on each later slice, warm-started from the previous slice’s weights.
- **RKHS**: kernel ridge with a Gaussian kernel of variance 0.1 in log-spot, 100 centres at the $j \cdot 100/101$ percentiles of the cloud, regularisation $\lambda = 10^{-9}$ (Bayer et al.’s published choices).
- **Ridge head on a frozen body**: the implicit network’s body (below) as a 65-dimensional feature map, with a per-slice ridge readout, $\lambda = 10^{-3}$ per sample, shrunk toward the previous slice’s coefficients.

3.3 Implicit loop

A single network $f_\theta(t, x)$, $2 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU with softplus output scaled by 0.04, inputs $(t/T, x/0.3)$. Starting from the explicit output as L^0 , each outer iteration simulates a full cloud under L^n with fresh random numbers, draws a pooled subsample of 60 000 points across all slices, trains θ for 300 Adam steps at learning rate 10^{-2} with the network and optimiser state carried over from the previous iteration, reads $\Phi(L^n)_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_\theta(t_k, \cdot), 10^{-4})}, 0, L_{\text{max}})$ on the fixed grid, and sets $L^{n+1} = (1 - \alpha)L^n + \alpha\Phi(L^n)$ with $\alpha = \frac{1}{2}$. The paper uses six iterations. The pooled data are truncated, v^+ , before training.

3.4 The variance-target discrepancy

Before 8 September 2026 the one-pass schemes handed the estimator the raw Euler v , while the implicit loop handed it v^+ . Since only v^+ drives the dynamics Equation (4), the quantity the leverage must divide by is $\mathbb{E}[V^+ \mid X]$. Under Feller violation a sizeable fraction of particles has $v < 0$ after a coarse step (on s11, $v_0 = 0.02$, $\xi = 1$, $\Delta = 0.04$: the one-step standard deviation of v is 0.028), so $\mathbb{E}[v \mid X] < \mathbb{E}[v^+ \mid X]$ by the mean negative excursion, the denominator was too small, and the leverage too large. The effect is independent of the particle count and shrinks with the step size. Fitting on v^+ (`ExplicitConfig.fit_v_floor`) has been the package default since 8 September 2026; every result in Sections 4 and 5 uses it, and the archived replays of the paper’s runs pin it off explicitly so that they still reproduce bit for bit.

4 Benchmark suite

This section specifies the clean benchmark agreed on 8 September 2026 and run on 9 September 2026 with the variance floor of Section 3.4 as the package default. Every cell of the tables below is one MLflow run, keyed by method, scenario, seed and, for the online stage, lag type and particle budget, in experiments `suite_cold`, `suite_offline`, `suite_lagged`, `suite_budget_tuned` and `pde_reference`. Every trained object is stored as a run artifact (network weights as a `torch` state dictionary, per-slice estimators and leverage fields as pickles and JSON) and can be reloaded by run id. The tables are generated from the store by `nparticle suite tables`.

4.1 Common setup

- **Scenarios.** The 20 SSVI scenarios of Table 1. The three Heston-market scenarios used in the literature comparisons (`li_simple`, `li_complex`, `bayer`; a Heston market surface, a different Heston variance process as the calibrated model, one-year maturity, log-strikes on $[-0.9, 0.9]$) are run in the same experiments and are in the store, but they score a single one-year maturity and are not reported in this draft.
- **Time grid.** 200 Euler steps for calibration and 200 for repricing, $\Delta = 0.01$ on $[0, 2]$, so that every method sits at its step-count plateau (the step-count diagnostics of 8 September 2026, MLflow experiment `mechanism`) and the comparison is about the estimator alone.
- **Particles.** 100 000 particles for every cold row and for the online stage at its full budget; at those sizes every per-slice fit and every online head sees the whole cloud, and the earlier code default of a 30 000-particle fit subsample is not used. The online budget is also swept below 100 000, and there too every fit uses the whole online cloud. Offline bodies at 200 000 and 500 000 particles: the fit subsample stays at 100 000, so each per-slice fit of a body reads a fresh 100 000-particle subsample of the offline cloud while the cloud itself is simulated at the full size.
- **Seeds.** Two seeds (0, 1) for every cold row and every online stage; one seed for each offline body. The reprice seed is 10 000 plus the calibration seed.
- **Floor.** All one-pass schemes fit $\mathbb{E}[V^+ | X]$; `fit_v_floor` becomes the default in this work and the tiny golden replays are regenerated in the same commit.
- **Implicit configuration.** Warm start from the explicit network field, 30 damped iterations at $\alpha = \frac{1}{2}$ with fresh noise per iteration, persistent network and optimiser (the best practical setting found in the convergence diagnostics of 8 September 2026, experiment `mechanism`).
- **Scoring.** As in Section 2.3: 500 000 repricing paths, Black–Scholes inversion at the snapped maturity, exact SSVI (or semi-analytic Heston) implied volatility as the target. Metrics per run: pooled and wings MAE and RMSE in volatility bp, both per maturity, the count of failed inversions, and the leverage RMSE against the PDE reference of the scenario. Errors are reported in implied-vol basis points; each table also shows the liquid-quote error (mean absolute error over $|k| \leq 0.25$ and $T \geq 0.5$) and the absolute option-price error in basis points of spot, recovered from the implied-vol errors through the Black–Scholes formula.
- **Latency.** Wall-clock seconds of the calibration only (repricing excluded). For the lagged suite, only the online stage is timed.
- **PDE reference.** The Fokker–Planck reference leverage (`reference/pde.py`) is solved per scenario at 200 steps and enters twice: repriced under the same Monte Carlo as every other row, which is the attainable floor of the scoring under this repricer (Gyöngy), and as the reference for the leverage-error column. Its repricing is repeated under a second reprice seed to measure the Monte Carlo noise floor of the score itself.

4.2 Cold suite

One calibration from scratch per method, scenario and seed at 100 000 particles. Methods, with their estimator of $\mathbb{E}[V^+ | X]$ per slice:

1. **NW:** Gaussian Nadaraya–Watson, Silverman bandwidth (Section 3.2).
2. **Explicit NN, short training:** per-slice network, warm-started across slices, 400 Adam steps on the first slice and 120 on each later slice, full batch (Section 3.2).
3. **Explicit NN, tuned:** the same per-slice scheme with a $1 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network, minibatch Adam (batches of 8192, learning rate 10^{-3}), 2000 steps on the first slice and 500 warm-started steps on each later slice; the architecture and schedule that Section 5 found

accurate in the wings and robust under Feller violation.

4. **Explicit NN, searched:** the per-slice scheme with the recipe found by the Optuna search of Section 5.4: a $1 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1$ SiLU network, minibatch Adam (batches of 8192, learning rate 3.7×10^{-3}), 500 steps on the first slice and 400 warm-started steps on each later slice, fitted on 250 000 particles of the cloud, with mean matching (the fitted slice is rescaled to reproduce the cloud’s mean variance), a monotone penalty on slopes of the wrong sign, and a linear continuation of the slice beyond its quantile grid.
5. **Implicit NN:** damped global-network loop (Section 3.3, configuration above).
6. **RKHS:** Gaussian kernel ridge with 100 quantile centres, Bayer et al.’s published knobs (Section 3.2).
7. **Spline:** penalised cubic B-spline, 25 quantile knots, $\lambda = 1$.
8. **Literature block.** GHL kernel: Nadaraya–Watson in spot with the quartic kernel and the Guyon–Henry–Labordère rule-of-thumb bandwidth $h(t) = 1.5 S_0 \sigma_{LV}(S_0, t) \sqrt{\max(t, 0.25)} N^{-1/5}$. Bins: piecewise-constant regression on 20 equal-frequency buckets (van der Stoep et al.). Muguruza: the kernel-free conditional Monte Carlo of Corollary 4.1, each particle contributing its one-step Gaussian conditional law of log-spot given the variance path. PU-RBF: Hakala’s partition-of-unity radial basis functions, 40 centres, 5-nearest-neighbour widths, regulariser 0.2.
9. **PDE:** the reference leverage repriced by Monte Carlo (attainable floor).

Table 2: Cold suite, 20 SSVI scenarios, mean and median over scenarios and two seeds. Liquid: $|k| \leq 0.25$, $T \geq 0.5$. The PDE row is the attainable floor of the scoring, its seed-to-seed spread is 6.6 bp (the Monte Carlo noise floor of the score), and its latency is the Fokker–Planck solve. Bold: the best method row in each column; the PDE floor is the reference and is not a contender.

method	implied vol, bp				price, bp	s
	mean	median	liquid	wings	pooled	median
NW	46	33	16	70	2.7	8.0
Explicit NN, short training	57	36	16	86	2.6	1042.7
Explicit NN, tuned	49	35	14	75	2.4	536.3
Explicit NN, searched	48	36	17	72	2.7	895.1
Implicit NN	58	37	19	88	3.3	1298.6
RKHS	81	70	45	113	8.5	12.4
Spline	47	34	14	73	2.5	2.7
GHL kernel	45	32	15	70	2.5	8.5
Bins	66	61	16	105	2.7	1.5
Muguruza	154	139	184	147	29.1	10.6
PU-RBF	57	40	16	90	2.8	2.5
PDE (attainable floor)	42	27	13	62	2.2	207.3

Results (Table 2). With the variance floor in place the classical kernels are the best cold calibrators: the GHL kernel and NW sit at 45–46 bp pooled, 3–4 bp above the attainable floor of 42, and the spline at 47. The tuned per-slice network closes most of the gap the short-trained network leaves ($57 \rightarrow 49$ bp pooled, $86 \rightarrow 75$ in the wings) and is the best row on liquid quotes (14 bp, against 15–16 for the kernels and 13 for the floor) and in price space (2.4 bp of spot against 2.5–2.7), but it does not beat the kernels on the pooled score: it is ahead of NW on 12 of the 20 scenarios, by a couple of basis points each, and behind by 14–39 bp on s03, s08 and s11, the scenarios where the variance spends most of its time at zero. Its latency is 536 s against 8 for NW, so as a cold calibrator it is a match, not an improvement. The searched network (Section 5.4) is 48 bp pooled, between NW and the hand-tuned network, better in the wings (72) and worse on liquid quotes (17), at 895 s: the search improved the estimator of $\mathbb{E}[V | X]$ by a few percent, and that does not change the cold picture. The implicit loop is 58 bp and the slowest row. Bins, PU-RBF, RKHS and Muguruza’s conditional Monte Carlo are behind the kernels by 10 to 110 bp, RKHS and Muguruza mostly in the

wings. Every row is dominated by the three-month wings; on liquid quotes all the competitive rows are within 3 bp of the floor, and in price space within 0.5 bp of spot.

4.3 What a lag is

An overnight calibration is done on surface S_0 ; the intraday market is S_1 , and the question is how much of the overnight work survives. Because the synthetic surfaces are parametric, a lag is a move in parameter space and never a bump of quoted points, so arbitrage cannot be introduced by accident. Two lags are run for every cell.

Surface lag. For SSVI, the existing bump: $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, shrunk by quarters until the Gatheral–Jacquier sufficient conditions ($\eta(1 + |\rho_s|) \leq 2$, $0 < \gamma \leq \frac{1}{2}$) hold; calendar no-arbitrage is automatic since $\theta_T = \sigma_0^2 T$ is increasing. For a Heston market, the matching move in vol terms: $\sqrt{v_0}$ and $\sqrt{\theta}$ up one volatility point, $\rho \rightarrow \min(\rho + 0.03, -0.05)$, $\xi \rightarrow 0.95\xi$; the surface is model-generated and arbitrage-free for any valid parameters.

Surface plus spot lag. The surface lag together with a sticky-strike spot move of +2%, $S_1 = S_0 e^\delta$, $\delta = \log 1.02$. Under sticky-strike the implied volatility at each absolute strike is unchanged, so in log-moneyness relative to the new spot the surface is translated: $\sigma'_{IV}(k, T) = \sigma_{IV}(k + \delta, T)$. A translation in k preserves Durrleman’s butterfly condition (it involves only w and its k -derivatives at a point) and calendar monotonicity, so the shifted surface is arbitrage-free. Call prices at fixed strike change with the spot, so the Dupire local volatility is that of the translated surface, $\tilde{w}(k) = w(k + \delta)$ with k measured from the new forward; the particles also start at the new spot. The stored denominators $f(t, x) = \mathbb{E}[V_t | X_t = x]$ were fitted for a cloud started at S_0 , and a conditional expectation given spot depends on the starting point: with $\rho < 0$ a particle sitting 2% above the old start had its variance drift down, so the stale f at the new spot is below v_0 , the stale leverage is too high near the money, most at short maturities. Sticky-delta, by contrast, leaves the surface in moneyness unchanged and the whole calibration is scale invariant, so a sticky-delta spot move is invisible to every method; the surface-only lag already covers it. A 2% daily move is one to one-and-a-half standard deviations at 20–30% volatility.

4.4 Offline bodies

Offline stage (experiment `suite_offline`, one run per scenario, body type and particle count): a full calibration on S_0 at 200 000 or 500 000 particles, 200 steps, seed 0. Three bodies are trained. The **explicit NN, short training** is the per-slice network pass on the paper’s schedule (400 Adam steps on the first slice, 120 on each later slice, full batch); the **explicit NN, tuned** is the same pass with the network and schedule of item 3 of Section 4.2 ($1 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU, minibatch Adam, 2000 steps on the first slice and 500 on each later slice); both save the per-slice networks and the per-slice denominators f_k . The **implicit NN** is the damped global loop of Section 3.3 and saves the global network. Each body is scored here as a cold calibrator on its own overnight surface S_0 , against the same repricer and the same quotes as Section 4.2, so the table says what a body is worth before any lag. These runs are cached and reused by every online cell below.

Results (Table 3). The tuned network is the only body worth keeping: 47 bp pooled at either size, against 54–55 for the short-trained network and 59–62 for the implicit loop, and it trains in half their time. More offline particles do not help any body at these training lengths (200k and 500k are within 1 bp), and the implicit loop gets slightly worse with more particles; the fit, not the cloud, is the limit. The tuned body’s liquid-quote error (14 bp) is within 1 bp of the floor. The searched body at 500 000 particles is 48 bp pooled, the best of the bodies in the wings (70) and the worst on liquid quotes (18); mean matching moves the near-money level while it fixes the wings. It trains in 856 s. (Only the 500 000-particle searched body was trained; it is the one the online head uses.)

Table 3: Offline bodies as cold calibrators on S_0 , 20 SSVI scenarios, seed 0; training time is the full offline calibration. Bold: the best method row in each column; the PDE floor is the reference and is not a contender.

body	particles	pooled	liquid	wings	price, bp	s
Explicit NN, short training	200k	54	15	83	2.6	1047.2
Explicit NN, short training	500k	55	16	84	2.7	1049.0
Explicit NN, tuned	200k	47	14	72	2.5	529.1
Explicit NN, tuned	500k	47	14	73	2.4	535.2
Explicit NN, searched	200k	—	—	—	—	—
Explicit NN, searched	500k	48	18	70	3.0	855.7
Implicit NN	200k	59	17	93	2.9	1365.5
Implicit NN	500k	62	20	95	3.3	1469.2
PDE (attainable floor)	—	42	13	62	2.2	207.3

4.5 Frozen body and online head

Online stage (one run per scenario, method, lag, online budget and online seed): given the intraday surface S_1 and a fresh cloud, build the intraday leverage and reprice. Only this stage is timed. The body is the tuned 500 000-particle body of Section 4.4, frozen; the methods differ in what is refitted online:

1. **Tuned body, stale f , fresh Dupire:** $L_k = \text{clip}(\sigma_{\text{Dup}}^{S_1}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\max})$ with the offline per-slice f_k . Costs nothing online beyond the repricing.
2. **Tuned body with spline head:** simulate the online particles under the stale- f leverage of item 1 with fresh Dupire, slice by slice; at each slice fit the penalised cubic B-spline of Section 3.2 (25 quantile knots, $\lambda = 1$) to the residual $v^+ - f_k(x)$ on the online cloud, and use $\max(f_k + \text{residual}, \frac{1}{4}f_k)$ as the denominator for the next step; the floor keeps the head from driving the leverage to its cap on Feller-violating scenarios.
3. **Tuned body with RKHS head:** the same residual head with a Gaussian kernel ridge (kernel width 0.5 times the cloud’s standard deviation, ridge 10^{-3}) in place of the spline, with the same floor.
4. **Tuned body with ridge head:** the per-slice network body frozen as a feature map (64 features plus a constant) with its last layer refitted by ridge on the online cloud, shrunk toward the offline readout. The body’s output is $\text{softplus}(Aw) \cdot 0.04$, and the refit minimises $\sum_i (\text{softplus}(A_i w) \cdot 0.04 - v_i^+)^2 + \lambda_{\text{eff}} \|w - w_0\|^2$ on the raw variance scale by Gauss–Newton to convergence (at most 20 steps) from the offline last layer w_0 , with $\lambda_{\text{eff}} = \lambda$ times the mean diagonal of the Gauss–Newton curvature at w_0 and $\lambda = 10^{-3}$, so the penalty is a fixed fraction of the data curvature whatever the output scale and cloud size. The denominator is $\text{softplus}(Aw) \cdot 0.04$ at the converged w . When the online data agree with the offline fit the first step is zero and $w = w_0$.

The online reference row is NW re-solved from scratch on S_1 at the same budget and 200 steps (the best intraday strategy in the floor-corrected bump runs, experiment `bump_floor`). The online budget is varied over 10 000, 30 000, 80 000 and 100 000 particles, with every fit on the whole online cloud: the 100 000 column is experiment `suite_lagged`, the other three are `suite_budget_tuned` (`scripts/budget_sweep.py`). The same four heads on the short-trained explicit body and on the implicit body are also in the store (experiment `suite_lagged`) and are summarised in Section 5.

Results (Table 4). A frozen tuned body with a guarded online head is at least as good as a fresh NW calibration at every online budget, and better where the budget is small. At 10 000 online particles NW re-solved from scratch is at 50 bp under both lags; the spline head is at 46, the RKHS and ridge heads at 46, and the advantage shrinks as the online cloud grows: at 80 000 particles, the practical budget, the spline head is 43 against NW’s 44–45 under both lags, at the same online time (7 s), and the ridge head is the best row after the spot move (43) at three times the time; at 100 000 every

Table 4: Frozen tuned or searched body (500 000 particles) with an online head, against NW re-solved from scratch, by online particle budget: pooled MAE in vol bp, 20 SSVI scenarios, two online seeds, both lags; liquid MAE at 100 000; online seconds at 10 000 and 100 000 particles (median). Bold: the best method row in each column; the PDE floor is the reference and is not a contender, and a zero online time (no online work) does not count.

method	surface lag					surface plus 2% spot lag					online s	
	10k	30k	80k	100k	liquid	10k	30k	80k	100k	liquid	10k	100k
NW re-solve on S_1	50	47	44	44	17	50	47	45	44	18	1.0	8.5
Tuned body, stale f , fresh Dupire	50	50	50	50	28	88	88	88	88	56	0.0	0.0
Tuned body + spline head	46	45	43	43	16	46	46	44	43	16	1.2	9.8
Searched body + spline head	46	45	42	42	16	46	45	43	43	16	1.0	7.1
Tuned body + RKHS head	46	46	44	44	16	46	46	44	44	16	2.3	18.8
Tuned body + ridge head	46	45	43	44	15	46	45	43	43	16	3.3	28.4

head is within 1–2 bp of NW and 1 bp of each other. On liquid quotes the heads are at 15–16 bp against NW’s 17–18 at every budget. The head is doing the work: the stale body with a fresh Dupire numerator is 50 bp under the surface lag and 88 after the spot move, at every budget, because the stored $\mathbb{E}[V | X]$ was fitted for a cloud started at the old spot. The spline head is the row to use: the cheapest head, never worse than the others, and it turns the tuned body’s cold-suite draw into a small but consistent win over the re-solve at every budget from 10k to 100k. The searched body with the spline head is the best row at every budget from 30 000 up: 42 bp under the surface lag at 80 000 and 100 000 against NW’s 44, and 43 after the spot move against 45 and 44, with the liquid-quote error at 16 bp against NW’s 17–18, and the fastest online sweep (7.1 s at 100 000 against 8.5 for NW), because the network is half the width. Its liquid-quote deficit as a cold calibrator disappears once the head runs on it. At 10 000 the searched and tuned bodies tie (46 bp). The search bought about half a basis point over the hand-tuned body at the practical budget and a faster head; the two-basis-point margin over the re-solve is the body-plus-head design, not the search.

4.6 Reproducibility

Each run records the git hash, the full configuration as parameters, the leverage field, the implied-volatility error grid, and the trained objects. A loader (`nparticle suite load <run_id>`) rebuilds the leverage field and, where present, the network or per-slice estimators from the artifacts alone, so any row of the tables can be re-scored or reused as an offline body without recomputation.

5 Follow-up studies: the estimator, the body, and the online budget

Three studies run on 10–11 September 2026 after the suite, on the six scenarios s01, s02, s05, s09, s11 and s16 (Feller ratios 1.5, 0.7, 0.3, 0.1, 0.08, 0.17). Every fit in these studies, and in the suite tables of Section 4, uses the full online cloud; the suite’s first pass, which fitted every per-slice estimator on the earlier code default of a 30 000-particle subsample, is kept in the store under the experiment names `suite*_sub30k` and is quoted below only where the comparison is the point. Scripts: `scripts/estimator_grid.py`, `scripts/conv_study.py`, `scripts/budget_sweep.py`; numbers from `scripts/study_tables.py`.

5.1 Estimator quality against the exact conditional expectation

Setup. For s02 (an easy surface) and s11 (Feller 0.08), a cloud of 500 000 particles is simulated under the PDE reference leverage of Section 4.1, and at the slices $t = 0.25$ and $t = 1$ each estimator is fitted on (x, v^+) and compared with the reference’s own conditional expectation f_{ref} on the resolved grid. The error is $|\hat{f} - f_{\text{ref}}|/f_{\text{ref}}$, averaged over the wings ($|x| > 0.25$) or the centre. Networks are the per-slice architecture of Section 3.2 with the hidden width and depth varied, trained from a cold start on 100 000 particles; “full batch” is Adam on the whole sample at learning rate 10^{-2} , “minibatch” is

Adam on batches of 8192 at 10^{-3} . NW is fitted on 30 000, 100 000 and 500 000 particles of the same cloud.

Conclusions. (i) The 30 000 subsample was costing real accuracy in the wings: on s02 at $t = 1$ NW’s wing error falls from 6.0% at 30k particles to 3.5% at 100k and 1.6% at 500k, while on s11 at $t = 0.25$ it is stuck at 17.5, 14.6 and 14.9%. (ii) On the easy surface the lever is training, not architecture: width 256 and depth 3 gain nothing, while the same 64×2 network on s02 at $t = 1$ goes 2.6% at the paper’s 120 full-batch steps, 1.3% at 1000 and 1.7% at 20 000 minibatch steps, better than NW on five times the particles and, over both slices, 0.6–1.7% in the wings for about two and a half minutes per slice; the 64×3 network at 20 000 minibatch steps sits alongside it at 1.8% (the warm start across slices in the real pass softens the short-training penalty). (iii) On the Feller-violating scenario every estimator, NW on 500k included, is 10–15% off the reference even at the money, and more particles do not help: the Euler cloud with full truncation at $\Delta = 0.01$ is not the continuous-time density, so the reference cannot judge estimators there. (iv) On that scenario the wing fit at $t = 0.25$ collapses to about 55% for every network trained for 120 full-batch steps, whatever its width or depth; the 3-layer networks escape from 1000 full-batch steps onward (12.8%, and 10.4% at 20 000 minibatch steps), the 2-layer networks only at 20 000 minibatch steps (55.1, 54.6, then 12.7%). Depth buys the escape earlier, not accuracy on easy surfaces.

5.2 Training the body

Setup. The offline bodies of Section 4.4 at 500 000 particles were retrained on the six scenarios under two training budgets and refitted on the full online cloud: “short” is the paper’s schedule (400 Adam steps on the first slice, 120 on each later slice; 300 steps per outer iteration for the implicit loop), “long” is 2000 and 1000 steps per slice (1000 per iteration on a 200 000-point pool for the implicit loop). Each body is scored as a cold calibrator on its own overnight surface, and the lagged suite is then run on it with every fit on the full 100 000-particle online cloud, alongside the first pass’s 30k-subsample rows.

Conclusions. (i) Training the explicit body ten times longer turns it from the weakest cold calibrator into the best one on these scenarios: pooled MAE 64 bp short with 30k fits, 70 bp short with 100k fits and 50 bp long, against NW’s 53, at 2.5 hours per body with full-batch training; the minibatch schedule of Section 5.1 should reach the same quality in a fraction of that and is the tuned body of Section 4.4. (ii) The implicit loop gets worse when trained harder (66, 78 and 88 bp along the same three arms) and its bodies are unstable across retrains; it is not a usable offline source. (iii) On the long-trained body at the full 100 000-particle online budget the heads sit on or just under the NW re-solve under both lags: NW 49 and 48 bp under the surface and the surface-plus-spot lag, the RKHS head 47 and 45, the spline head 48 and 49, the ridge head 48 and 47, while the stale body alone is 61 and 99 and so remains useless after the spot move. (iv) The RKHS head of the first pass had no positivity guard and a fixed kernel width, and drove the denominator to its floor on the Feller-violating scenarios (s09: 858 bp on the long-trained body); the guard and the adaptive width of Section 4.5 were introduced here, and the suite’s RKHS rows were recomputed with them.

5.3 The online budget

Setup. The long-trained explicit body at 500 000 particles with each head, against NW re-solved from scratch, at online budgets of 10 000, 30 000 and 80 000 particles (every fit on the whole cloud), both lags, six scenarios, two seeds; the 100 000 column is the lagged suite of Section 5.2. Experiment `suite_budget`.

Conclusions. At 100 000 particles every method sits within a few basis points of the floor and the comparison cannot separate them. As the online cloud thins under the surface lag, NW degrades, 49, 50, 52 and 57 bp at 100k, 80k, 30k and 10k, and the body with a head does not: the RKHS head is 47, 47, 48 and 50, the spline head 48, 48, 50 and 51. At 10 000 particles the RKHS head is 7 bp ahead under both lags, 9 bp in the wings and 6 bp at two years, and the body with either kernel

head at 10k matches NW at 80k with eight times fewer particles; at that budget the online stage costs 1.0 s for NW, 0.9 s for the spline head, 2.1 s for the RKHS head and 2.9 s for the ridge head, against nothing for the stale body. The head is doing the work, not the body: the stale body alone is flat at 61 bp and useless after the spot move at every budget. The ridge head is the weakest at 10k (54 bp), where it refits 65 features on the thinnest data.

5.4 Searching the estimator recipe

Setup. On 12 September 2026 the per-slice estimator’s recipe was searched with Optuna (experiment `optuna_offline`, study `offline_v1`, one nested MLflow run per trial). The search space: hidden width $\{32, 64, 128\}$, depth $\{2, 3, 4\}$, learning rate in $[10^{-4}, 10^{-2}]$, batch $\{2048, 8192, \text{full}\}$, first-slice steps in $[500, 4000]$, later-slice steps in $[100, 1500]$, weight decay, fit sample $\{100, 250, 400\} \times 10^3$, warm start on or off, and four structural priors that can be switched on: mean matching (the fitted slice rescaled to the cloud’s mean variance), a monotone penalty $\lambda \mathbb{E}[\text{relu}(-\text{sign}(\rho) \partial_x f)^2]$, a heteroscedastic loss weighted by the inverse local variance, and the continuation of the slice beyond its quantile grid (flat, linear, or the network itself). The search never sees the 20 reported scenarios: 24 fresh SSVI scenarios were drawn from the registry’s parameter ranges, stratified so that a third have Feller ratio below 0.3, and for each a cloud of 500 000 particles was simulated once under the NW-calibrated leverage and its slices at $t \in \{0.1, 0.5, 1, 1.75\}$ stored, split 80/20 into a fit pool and a held-out set. A trial fits the four slices of eight scenarios in the warm-started chain and is scored by the log of its held-out squared error relative to NW’s on the same particles, averaged over slices and scenarios, so that 0 is “as good as NW” and negative is better; a median pruner stops bad trials after two scenarios and a 15-minute cap stops slow ones. 200 trials on four workers took three hours: 118 complete, 73 pruned, 9 timed out.

What the score can resolve. The held-out squared error is the estimation error of f plus the conditional variance of V given X , which no estimator can remove; on these clouds the second part is 99.4% of the total, so NW’s own estimation error is 0.6% of its loss and the scores cluster within a few tenths of a percent of zero. The best trials reached -0.0016 , about 0.16% lower total error than NW, which is roughly 5–10% lower estimation error; converting the best trials’ per-slice ratios with the estimated conditional variance gives medians of 0.92–0.98 across slices. The ranking is unaffected by the constant, but the comparison is noisy at the scale of the effect, which is why the seeds are fixed per scenario and the top recipes are validated on the repricing score.

What the search found. Every top recipe has width 32, minibatch training, mean matching on, the monotone penalty on (at a weight so small it acts as a tie-breaker), the linear tail, warm start on, the heteroscedastic loss off, and a fit sample of 250 000; depth and schedule are interchangeable. The winner (trial 189): depth 4, batches of 8192, learning rate 3.7×10^{-3} , 500 then 400 steps, the cheapest of the finalists.

Validation. The top three recipes were each trained as offline bodies at 500 000 particles on the 20 reported scenarios and run with the spline head at 80 000 online particles under both lags against NW re-solved at 80 000 (pooled MAE, surface lag / surface plus spot lag): NW 44.4/44.8, the hand-tuned body 42.8/43.5, trial 189 42.4/43.0, trial 191 42.7/43.2, trial 178 42.4/43.1; liquid-quote error 16.0 for every body against 17.7/18.0 for NW. Trial 189 was promoted as the searched recipe (`estimators/recipes/explicit_opt.json`) and appears as “searched” in Tables 2 to 4: as a cold calibrator it is a draw with the kernels like the hand-tuned network; as a frozen body with the head it is the best row at every budget from 30 000 up, by half a basis point over the hand-tuned body and two over the re-solve, with the fastest online sweep.

5.5 What the studies say

With the variance floor in place, the classical one-pass kernels are the best cold calibrators and sit a few basis points above the scheme’s own floor, so at 100 000 particles no estimator can look better than “a few basis points”. A per-slice network is only competitive once it is trained properly,

and even a searched recipe does not beat the kernels as a cold calibrator at 100 000 particles: the estimator of $\mathbb{E}[V \mid X]$ can be made a few percent better than NW, and that is worth basis points only through a frozen body and an online head. A frozen, properly trained body with an online head reprises today's vanillas at least as well as a fresh NW calibration at any online budget, and clearly better when the online budget is small, which is where the latency argument for the head lives. The implicit loop is not a useful offline source. The remaining distance to a 10 bp tolerance on liquid quotes is the scheme's, not the estimator's.